

145 mm.

THYROSIT 100 mcg TABLET

DESCRIPTION

White round biconvex tablet with a bisect and embossed with "THYROSIT", "SPS" on one side, plain on the other. Each tablet contains L-Thyroxine sodium 100 mcg.

PHARMACODYNAMICS

Thyroxine is an essential hormone that affects the rate of many physiologic processes especially the metabolic rate of body tissues and the regulation of cell growth and differentiation. Thyroxine will reverse the signs and symptoms of hypothyroidism and myxoedema by increasing renal blood flow and glomerular filtration rate, resulting in a diuresis within 24 hours following administration. In hypothyroid children, thyroxine increases epiphyseal growth and bone ossification. It is used to reduce goitre size by suppressing the secretion of thyrotropin.

PHARMACOKINETICS

Following oral administration, thyroxine sodium is absorbed from the GI tract. The amount absorbed increases during fasting conditions. It is distributed into most body tissues and fluids with highest concentrations in the liver and kidneys. Thyroxine does not readily cross the placenta and minimal amounts are secreted into milk. About one third of absorbed thyroxine (T4) is converted, by deiodination, to triiodothyronine (T3), an active metabolite. Thyroxine is more than 99% bound to serum proteins resulting in high serum concentration and slow metabolic clearance. The plasma half-life is 6-7 days and has the advantage that it needs only be given as a single daily dose. About 20-40% of thyroxine is eliminated unchanged in the faeces while 85% is metabolized daily through deiodination. Iodine liberated by deiodination reaction is utilised by thyroid gland for hormone synthesis or is excreted in faeces via bile or in urine.

INDICATIONS

Thyrosit 100 mcg is used in the:

- Treatment of hypothyroidism.
- Replacement or substitution of diminished or absent thyroid function resulting from primary causes including functional deficiency, primary atrophy and partial or complete absence of the gland or from the effects or surgery, radiation of thyroid glands.
- Replacement or supplemental therapy in patients with secondary or tertiary hypothyroidism.
- Treatment of congenital hypothyroidism (cretinism).
- Replacement therapy in the management of simple goitre and chronic lymphocytic thyroiditis.

DOSEAGE AND ADMINISTRATION

The dose of Thyrosit 100 mcg for the treatment of any thyroid disorder should be individualised on the basis of clinical response and biochemical tests and monitored regularly.

The usual adult maintenance dose is 100-200 mcg daily. For mild hypothyroidism in adults, the initial dose is 50 mcg daily and is increased by 25-50 mcg daily at intervals of 2-4 weeks until the desired response is obtained. In healthy adults with relatively recent onset of hypothyroidism, 100-200 mcg daily is given. For severe hypothyroidism in adult, the initial dose is 12.5-25 mcg daily and increased by 25 mcg daily at intervals of 4 weeks.

In elderly patients, in patients with coronary heart disease, and in patients with severe or long-existing hypothyroidism, special caution is required when initiating therapy with thyroid hormones, that is, a low initial dose (for example 12.5 mcg/day) should be given which should then be increased slowly and at lengthy intervals (e.g. a gradual increment of 12.5 mcg/day fortnightly) with frequent monitoring of thyroid hormones. A dosage, lower than optimal dosage giving complete replacement therapy, consequentially not resulting in a complete correction of TSH level, might therefore need to be considered.

In children, individualisation of dosage and monitoring of treatment is especially important. It is essential to achieve rapid and complete thyroid replacement, hence a dosage should be given until mild toxic symptoms appear, and then the dose is reduced slightly. For neonates and infants with congenital hypothyroidism, where rapid replacement is important, the initial recommended dose is 10 to 15 mcg per kg BW per day for the first 3 months. Thereafter, the dose should be adjusted individually according to the clinical findings and thyroid hormone and TSH values.

Dosing guideline for paediatric hypothyroidism are as the following:

Age	mcg/kg/dose, daily
0 - 3 months	10 - 15
3 - 6 months	8 - 10
6 - 12 months	6 - 8
1 - 5 years	5 - 6
6 - 12 years	4 - 5
> 12 years	2 - 3

The daily doses can be given in single administration. Ingestion: as a single daily dose in the morning on an empty stomach, half an hour before breakfast, preferably with a little liquid, (for example, half a glass of water). Infants receive the entire dose at once at least 30 minutes before the first meal of the day. Tablets are to be disintegrated in some water and the resultant suspension, which must be prepared freshly as required, is to be administered with some more liquid. Duration of treatment is usually for life in case of substitution in hypothyroidism and after strumectomy or thyroidectomy and for relapse prophylaxis after euthyroid goitre removal. Concomitant therapy of hyperthyroidism after achieving euthyroid status is indicated for the period in which the anti-thyroid drug is given. For benign euthyroid goitre, a treatment duration of 6 months up to 2 years is necessary. If medical treatment was not sufficient within this time, surgery or radioiodine therapy of the goitre should be considered.

ROUTE OF ADMINISTRATION

Oral

CONTRAINDICATIONS

Thyrosit 100 mcg is generally contraindicated in the following conditions:

- Thyrotoxicosis.
- Acute myocardial infarction uncomplicated by hypothyroidism.
- Uncorrected adrenal insufficiency.
- Untreated subclinical (suppressed serum TSH level with normal T3 and T4 levels) hyperthyroidism.
- Known hypersensitivity to any component in this preparation.

WARNINGS AND PRECAUTIONS

Patients receiving thyroid agents must be closely monitored and replacement therapy should be initiated with low doses.

Levothyroxine should be used with extreme caution in patients with cardiovascular disorders, including hypertension, angina, coronary artery disease, heart failure, myocardial infarction, and in the elderly who have a greater likelihood of occult cardiac disease. Cardiac disease may be exacerbated by the administration of thyroid hormones, resulting in angina pectoris, myocardial infarction or sudden cardiac death. An electrocardiogram (ECG) performed before starting treatment with levothyroxine may help to distinguish underlying myocardial ischaemia from changes induced by hypothyroidism. In the presence of cardiac disease, dosage should be started cautiously and increased slowly.

Patients with diabetes mellitus receiving antidiabetic agents (i.e. insulin, sulphonylureas) concomitantly should be advised to monitor closely urinary and/or blood glucose concentrations.

Children receiving a thyroid agent should be advised that partial loss of hair may occur during the first few months of therapy, but this effect is usually transient and subsequent regrowth usually occurs.

Patients undergoing surgery, fasting and with abnormalities in thyroid function tests observed in nonthyroidal illness may be suffering from Euthyroid Sick Syndrome (ESS). ESS is described as abnormal findings on thyroid function tests that occur in the setting of a nonthyroidal illness (NTI) without pre-existing hypothalamic-pituitary and thyroid gland dysfunction.

Patients with rare hereditary problems of galactose intolerance, the Lapp lactose deficiency or glucose-galactose malabsorption should not take this medicine.

Patients with panhypopituitarism or other disorders predisposing to adrenal insufficiency may react to levothyroxine treatment, and the adrenal insufficiency should be corrected with corticosteroid therapy before commencing treatment with levothyroxine, to prevent precipitation of acute adrenal crisis.

Some patients may or may not respond adequately to Thyrosit 100 mcg Tablet due to one or more of the following reported reasons:

- a) Low serum T3 due to starvation and stress
Patients undergoing a period of stress and starvation may induce conversion of T4 into rT3 (inactive and reversed form of T3) instead of the active T3. This is because, starvation or carbohydrate deprivation rapidly inhibits deiodination of T4 to T3 by type 1 5'- monodeiodinase (D1) in the liver, thus inhibiting the generation of T3 and preventing the metabolism of rT3. Consequently, there is a drop in serum T3 and an elevation in rT3 except in renal failure patients.

High levels of glucocorticoids are known to suppress the pituitary response to thyrotropin-releasing hormone (TRH) in man and stress-induced elevation of glucocorticoids may cause suppression of TRH-pituitary response, decrease in TSH, serum T4 and T3 levels.
- b) Impairment of 5'-monodeiodinase activity
Two peripheral enzymes deiodinate T4: 5'- monodeiodinase converts T4 to active T3 while 5- monodeiodinase converts T4 to inactive rT3. Some patients with mutations or genetic polymorphisms in the deiodinase enzymes (specifically type 1 5'- monodeiodinase (D1) and type 2 5'- monodeiodinase (D2)) may affect the regulating serum T3 levels. It has been shown that decreased deiodinase activity may contribute to dysregulation of serum T3 levels.
- c) Illness
Euthyroid patients with chronic illnesses, trauma or surgery may have low T3 syndrome – that is low serum T3 levels in the face of normal TSH and T4. Cardiac cells are dependent on circulating serum T3 with many congestive heart failure patients having low T3 syndrome.
- d) Increased cytokines level
Current studies have shown that cytokines may induce a condition that is associated with low supplies of serum T4 and T3, low production levels of T4 and T3 and low tissue levels of T4 and T3. This condition of low T3 and T4 is due to impairment in 5'- monodeiodinase activity in the increased presence of cytokines. Treatment with T4 monotherapy will not improve the serum T3 level without substantially increase T4 level beyond normal due to impaired conversion.
- e) Drug interactions
Dopamine and corticosteroids - suppress pituitary TSH release and inhibit conversion of T4 to T3 resulting in reduced level of total T3, free-T4 and TSH in patients.

Amiodarone / iodinated radiocontrast agents – increase level of total and free-T4 in patients.
Intravenous / subcutaneous heparin therapy – increases free-T4 level due to in vitro interference with laboratory assay. Patients may remain clinically euthyroid with normal total T4 and TSH levels.

High serum T4 is also seen in patients with elevated serum concentration of thyroxine-binding globulin (TBG). High serum TBG is also elevated in patients with acute intermittent porphyria and several liver diseases, including chronic hepatitis and primary biliary cirrhosis.
- f) Low serum T4
Patients with serious illness, reduced T3 and T4 concentrations are associated with increased severity and length of illness with mortality rate directly correlated to serum T4 concentration.

Monotherapy with T4 may fail to promptly increase T3 concentration due to the inability of these patients to convert T4 to T3. To restore T4 and T3 concentration to normal level, such patients should be treated with a combination of T4 and T3.

In patients with ESS, serum T4 is commonly below normal level, but free T4 have been reported to be at normal or above normal levels. This variability in free T4 levels reflects both the assay method used in testing and the patient's underlying illness. The result of free T4 assays can be influenced by the presence of circulating inhibitor of serum binding hormones, effects of drugs, metabolites or free fatty acids in the serum or blood assay.

- g) Thyrotropin-Releasing Hormone (TRH) and Thyroid-Stimulating Hormone (TSH)
Low TRH secretion in hypothalamus causes low output of pituitary TSH and thyroid hormones (T4 and T3). Patients having low TSH level but respond normally to T4 replacement therapy may suggest impairment of hypothalamic function.
- h) Uneven effects of T4 replacement
Under normal physiological conditions, majority of circulating T3 is derived from peripheral deiodination of T4 with approximately 20% of T3 directly secreted by thyroid gland. In replacement therapy, all the T3 is derived from peripheral conversion of T4 with excess of T4 being required to normalized TSH. Patients may be euthyroid as defined by circulating thyroid hormone levels and particularly TSH, but symptoms may arise from uneven effects of T4 replacement therapy on various tissues and organs or even regions within the brain.
- i) Abnormal Thyroid Hormone Binding Proteins (THBPs)
Certain patients have abnormalities in the thyroid test due to genetic variation (abnormal hormone binding proteins). In blood, circulating T4 and T3 are bound to THBPs such as thyroxine binding globulin (TBG), serum albumin and transthyretin (TTR). This serves as an extrathyroidal reservoir of the hormones and to ensure an uninterrupted supply of thyroid hormones is available to various tissues and organs.

Any alteration in the concentration of THBPs or their affinities for thyroid hormones may cause significant changes in the binding capacities leading to changes in ratio of free to bound thyroid hormone concentration in circulation. This altered thyroid hormone levels caused by THBP abnormalities may create confusion in the clinical diagnosis of thyroid function.
- j) Effects on bone mineral density
In women, long-term levothyroxine sodium therapy has been associated with increased bone resorption, thereby decreasing bone mineral density, especially in post-menopausal women on greater than replacement doses or in women who are receiving suppressive doses of levothyroxine sodium. The increased bone resorption may be associated with increased serum levels and urinary excretion of calcium and phosphorus, elevations in bone alkaline phosphate and suppressed serum parathyroid hormone levels. Therefore, it is recommended that patients receiving levothyroxine sodium should be given the dose necessary to achieve and desired clinical and biochemical response.
- k) Selenium deficiency
Selenium plays an essential role in the metabolism of thyroid hormone. Studies have shown that selenium deficiency decreases the synthesis of thyroid hormones leading to hypothyroid symptoms,as it decreases the function of iodothyronine deiodinases (selenoprotein) that regulate the thyroid hormone turnover and catalyse the conversion of T4 to its biologically active form, T3.
- L) Brand switching
If the brand of levothyroxine medication is changed,either from one brand to another brand, from a brand to a generic brand, or from a generic brand to another generic brand, patients should be re-tested by measuring serum TSH and free T4 in six (6) weeks, and readjust the dose as needed. Since small changes in levothyroxine administration can cause significant changes in TSH serum concentrations, precise and accurate TSH control is necessary to avoid potential adverse iatrogenic effects. This is of particular concern in frail patients,those with thyroid cancer, and the paediatric age group. It is recommended that patients be started and maintained on either brand-name or generic levothyroxine preparations and not switched back and forth between the two.

DRUGS INTERACTIONS AND FOOD EFFECTS

Many drugs affect thyroid hormone pharmacokinetics and metabolism (e.g. absorption, synthesis, secretion, catabolism, protein binding, and target tissue response) and may alter the therapeutic response to levothyroxine.

The magnitude of the effects is likely to be patient-specific and may vary by such factors as age, gender, race, intercurrent illnesses, dose of either agent, additional concomitant medications, and timing of drug administration.

Numerous factors, alone or in combination, including diet, other medications, botanicals may influence response of patient to levothyroxine.

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145 mm.

Interactions affecting other drugs:

Antidiabetic agents:
Levothyroxine may reduce the effect of antidiabetic agents. For this reason, blood glucose levels should be checked frequently at the start of thyroid hormone therapy and the dosage of antidiabetic agents has to be adapted, if necessary.

Coumarin derivatives:
The effect of anti-coagulant therapy can be intensified as levothyroxine displaces anti-coagulative drugs from plasma proteins, which may increase the risk of haemorrhage, e.g. CNS or gastrointestinal bleeding, especially in elderly patients. Therefore, it is necessary for coagulation parameters to be checked regularly at the start of and during concomitant therapy. If necessary, the dosage of the anti-coagulative drug has to be adapted.

Antidepressant:
Antidepressant response to tricyclics (e.g. amitriptyline, imipramine and dosulepin) can be accelerated because levothyroxine increases sensitivity to catecholamine; concomitant administration may precipitate cardiac arrhythmias.

Anti-inflammatory treatments:
False low plasma concentration has been observed with concurrent anti-inflammatory treatment such as phenylbutazone or acetylsalicylic acid and levothyroxine therapy.

Beta-blockers:
Levothyroxine (thyroxine) accelerates metabolism of propranolol, atenolol and sotalol.

Cardiac glycosides:
If levothyroxine therapy is initiated in digitalized patients, the dose of digitals may require adjustment. Hyperthyroid patients may need their digoxin dosage gradually increased as treatment proceeds because initially patients are relatively sensitive to digoxin.

Sympathomimetic:
Effects of sympathomimetic agents (e.g. adrenaline or phenylephrine) may be enhanced by levothyroxine.

Ketamine:
Cautious administration of ketamine is recommended in patients on levothyroxine therapy, as marked hypertension and tachycardia may occur.

Interactions affecting levothyroxine:
Aluminium-containing drugs, iron-containing drugs, calcium carbonate: Aluminium-containing drugs (antacids, sucralfate) have been reported in the pertinent literature as potentially decreasing the effect of levothyroxine. Drugs containing levothyroxine should therefore be administered at least 2 hours prior to the administration of aluminium-containing drugs.Same is true for iron-containing drugs and calcium carbonate.

Ion-exchange resins
(e.g. cholestyramine, sodium polystyrene sulphonate or colestipol):
Due to concurrent use of ion-exchange resins with levothyroxine, the effects of levothyroxine may be reduced due to the resin binding to levothyroxine in the gastrointestinal tract, causing a delay or impairment in levothyroxine absorption. An interval of 4 to 5 hours between the administrations of the two medications is recommended.

Protease inhibitors:
Protease inhibitors (e.g. ritonavir, indinavir, lopinavir) may influence the effect of levothyroxine. Close monitoring of thyroid hormone parameters is recommended. If necessary, the levothyroxine dose has to be adjusted.

Sevelamer:
Sevelamer may decrease levothyroxine absorption. Therefore, it is recommended that patients are monitored for changes in thyroid function as the start or end of concomitant treatment. If necessary, the levothyroxine dose has to be adjusted.

Tyrosine kinase inhibitors:
Tyrosine kinase inhibitors (e.g. imatinib, sunitinib) may decrease the efficacy of levothyroxine. Therefore, it is recommended that patients are monitored for changes in thyroid function at the start or end of concomitant treatment. If necessary, the levothyroxine dose has to be adjusted.

Antiepileptic:
Carbamazepine, barbiturates including phenobarbital, phenytoin and primidone can accelerate levothyroxine metabolism. Initiation or discontinuation of anticonvulsant therapy may alter levothyroxine dosage requirements.

Sertraline, chloroquine/proguanil:
These substances decrease the efficacy of levothyroxine and increase the serum TSH level.

Antibacterial:
Rifampicin enhances thyroid hormone metabolism which may increase requirements for levothyroxine in hypothyroidism. Oral ciprofloxacin may decrease the absorption of levothyroxine. An interval of 6 hours between the administrations of the two medications is recommended.

Sex hormones:
Oestrogen, oestrogen-containing products (including hormone replacement therapy) and oral contraceptives may increase the requirement of thyroid therapy dosage. Conversely, androgens may decrease serum concentrations of levothyroxine-binding globulins.

Propylthiouracil, glucocorticoids, beta-sympatholytic, amiodarone and iodine-containing contrast media:These substances inhibit the peripheral conversion of T4 to T3.

Proton pump inhibitors: Absorption of levothyroxine is possibly reduced by proton pump inhibitors. Administration should be separated by 4 to 5 hours.

Ulcer healing drugs: Sucralfate or cimetidine may reduce the absorption of levothyroxine. Administration should be separated by 4 to 5 hours.

Weight loss drugs:
Orlistat may decrease levothyroxine absorption which may result in hypothyroidism (monitor for changes in thyroid function).

Food:
Goitrogenic foods (asparagus, cabbage, peas, turnip greens, broccoli, spinach, brussels sprouts, lettuce, soybeans, soybean flour, cottonseed meal, walnuts, dietary fibre) decrease absorption of levothyroxine from GI tract.

Soy-containing compounds:
Soy-containing compounds can decrease the intestinal absorption of levothyroxine. Therefore, a dosage adjustment of thyroxine may be necessary, in particular at the beginning or after termination of nutrition with soy supplements.

Others:
Binding to Serum Proteins: The following agents may either inhibit levothyroxine sodium binding to serum proteins or alter the concentrations of serum binding proteins: androgens and related anabolic hormones, asparaginase, dicumarol, clofibrate, oestrogens and oestrogen-containing compounds, 5-fluorouracil, furosemide, glucocorticoids, meclofenamic acid, mefenamic acid, methadone, perphenazine, phenylbutazone, phenytoin, salicylates, tamoxifen.

Thyroid Physiology: The following agents may alter thyroid hormone or TSH levels, generally by effects on thyroid hormone synthesis, secretion, distribution, metabolism, hormone action, or elimination, or altered TSH secretion: aminoglutethimide, p-aminosalicylic acid, amiodarone, androgens and related anabolic hormones, complex anions (thiocyanate, perchlorate, pertechnetate), antithyroid drugs, β-adrenergic blocking agents, carbamazepine, chloral hydrate, diazepam, dopamine and dopamine agonists, ethionamide, glucocorticoids, heparin, hepatic enzyme inducers, insulin, iodinated cholecystography agents, iodine-containing compounds, levodopa, lovastatin, lithium, 6-mercaptopurine, metoclopramide, mitotane, nitroprusside, phenobarbital, phenytoin, resorcinol, rifampin, somatostatin analogs, sulfonamides, sulfonyleureas, thiazide diuretics.

Adrenocorticoids: Metabolic clearance of adrenocorticoids decreased in hypothyroid patients and increased in hyperthyroid patients and may therefore change with changing thyroid status.

Risk-benefit should be considered when the following medical problems exist: Adrenocortical insufficiency must be corrected while thyroid replacement therapy is being given, to prevent precipitation of acute adrenocortical insufficiency.

Cardiovascular disease including angina pectoris, arteriosclerosis, coronary artery disease, hypertension, myocardial infarction (because of the risks associated with overly rapid thyroid hormone replacement and increased metabolic demands; mobilization of myxoedema fluid may produce pitting oedema 1 to 3 or more weeks after change in dosage).

Diabetes mellitus possible reduced glucose tolerance and increased insulin or oral antidiabetic agent requirements.

Hyperthyroidism - history of residual autonomous thyroid function may be present after therapy for hyperthyroidism, necessitating lower than typical doses.

Malabsorption states, such as celiac disease - absorption, especially of levothyroxine, is reduced; dosage adjustment may be necessary.

Pituitary insufficiency - associated adrenocortical insufficiency must be corrected before thyroid replacement therapy is initiated, to prevent precipitation of acute adrenocortical insufficiency.

Sensitivity to thyroid hormone - Thyrotoxicosis being treated with antithyroid medication, caution is required also in patients with long-standing hypothyroidism or myxoedema, who may be more sensitive to effects of thyroid hormones.

Patient monitoring - in patients receiving levothyroxine, liotrix or thyroid extract for primary hypothyroidism, serum free T4 index (total serum T4 and T3 resin uptake) or serum free T4 together with a serum thyroid-stimulating hormone (TSH) are the most useful tests for monitoring replacement therapy.

General Dosing Information - In most cases, relative need for thyroid hormone dosage adjustment will depend on the thyroid state of the patient and the dosages of all medications involved.Dosage adjustment should be based on results of thyroid function tests and clinical status. Inadequate doses of levothyroxine sodium may produce or fail to resolve symptoms of hypothyroidism.

PREGNANCY AND LACTATION

Pregnancy
The safety of levothyroxine during pregnancy has not been established. However, pregnancy occurring in a woman on a maintenance dose of levothyroxine for thyroid deficiency would be expected to proceed normally; However, maternal levothyroxine doses should be monitored as requirements may change during pregnancy. Thyroid hormones appear not to readily cross the placenta. The risk of foetal abnormalities should be weighed against that of untreated hypothyroidism.

Breastfeeding

The concentration of thyroid hormones excreted into breast milk appears not to be clinically significant but screening for congenital hypothyroidism might be affected.

SIDE EFFECTS

Adverse reactions associated with levothyroxine therapy are primary those of hyperthyroidism due to therapeutic overdose. The adverse reactions other than those indicative of thyrotoxicosis, either initially or during the maintenance periods, are rare.

Adverse reactions include the following:

General: fatigue, increased appetite, weight loss, heat intolerance, fever, excessive sweating.

Central nervous system: headache, hyperactivity, restlessness, nervousness, anxiety, irritability, emotional lability, insomnia.

Musculoskeletal: tremors, muscle weakness; cramps in skeletal muscle, craniostynosis in infants and premature closure of the epiphyses in children.

Cardiovascular: flushing, palpitations, tachycardia, arrhythmias, increased pulse and blood pressure, heart failure, angina, myocardial infarction, cardiac arrest.

Respiratory: dyspnoea.

GI: diarrhoea, vomiting, and abdominal cramps.

Hepatic: liver function test increased.

Dermatologic: hair loss; partial loss of hair may occur in children during the first few months of therapy but this effect is transient and subsequent regrowth usually occurs. Skin reactions have been reported rarely.

Reproductive: menstrual irregularities, infertility. Pseudotumor cerebri (benign intracranial hypertension) has been reported in children receiving levothyroxine therapy. Seizures have been reported rarely with the institution of levothyroxine therapy.

Immune system disorders: hypersensitivity reactions (rare) including rash, pruritus, dyspnoea, joint pain, malaise, eosinophilia, liver dysfunction, urticaria and oedema.

Psychiatric disorders: mania

Metabolic: fever, glucose intolerance, weight loss, premature craniostynosis (in children) TRH suppression, heat intolerance, sweating, flushing.

Some patients may experience a severe reaction to high levels of thyroid hormone. This is called a "thyroid crisis" with any of the following symptoms: Hyperpyrexia, tachycardia, arrhythmia, hypotension, cardiac failure, jaundice, confusion, seizure and coma.

OVERDOSE AND TREATMENT

Symptoms of overdose: Tachycardia/arrhythmia, palpitations, angina pain, headache, nervousness, restlessness, confusion, anxiety, hypertension, excitability, sweating, flushing, tremor, fever, weight loss, muscle weakness and cramps, menstrual irregularities, diarrhoea and vomiting. These signs can take up to 5 days to appear. Atrial fibrillation may develop. There may be increased toxicity in those with pre-existing heart disease.

Treatment: In the treatment of acute thyroid agent overdose, symptomatic and supportive therapy should be instituted immediately. Treatment consists principally of reducing GI absorption of the drugs and counteracting central and peripheral effects, mainly those of increased sympathetic activity. Initial treatments include induced emesis, gastric lavage and ingestion of activated charcoal. If the patient is comatose, having seizures or lacks the gag reflex, gastric lavage may be performed if an endotracheal tube with cuff inflated is in place to prevent aspiration of gastric contents. Oxygen may be administered and ventilation maintained. If congestive heart failure develops, cardiac glycosides may be administered. Measures to control fever, hypoglycaemia or fluid loss should be initiated as necessary. β-adrenergic blocking agents (e.g. propranolol) are useful to counteract many of the effects of increased sympathetic activity. Propranolol may be given intravenously at a dosage of 1 mg to 3 mg over a 10 minutes period or orally at 80 mg to 160 mg per day, especially when no contraindication exists for its use. Other adjunctive measures may include administration of cholestyramine to interfere with thyroxine absorption, and glucocorticoid to inhibit conversion of T4 to T3.

PACKING
Amber-PVC-Alu-blisters pack of 10's. Box of 10, 20 and 50 strips.

STORAGE CONDITIONS & PHARMACEUTICAL PRECAUTIONS

Store in a cool dry place, below 30°C.

Protect from light.

Keep out of reach of children.

(Jaui dari pada anak-anak)

Shelf-life: Please refer to the outer box label

Manufactured by:

Sripasit Pharma Co., Ltd.
216 Moo 6, Suanluang,
Kratum Baen,
Samut Sakhon 74110, Thailand.

Revision date: Jan 2021

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PIT0003 Rev: 7

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